

Second-Order Linear Homogeneous Equations

ANSWER KEY



The auxiliary equation

- 1 Write the auxiliary (characteristic) equation for $y'' - 5y' + 6y = 0$.
 $r^2 - 5r + 6 = 0$.
- 2 Write and simplify the auxiliary equation for $2y'' + 4y' - 6y = 0$.
 $2r^2 + 4r - 6 = 0$, which simplifies to $r^2 + 2r - 3 = 0$.

Distinct real roots

- 3 Solve $y'' - 5y' + 6y = 0$.
 $r^2 - 5r + 6 = (r - 2)(r - 3) = 0 \Rightarrow r = 2, 3$. General solution: $y = C_1e^{2x} + C_2e^{3x}$.
- 4 Solve $y'' + 2y' - 3y = 0$, $y(0) = 1$, $y'(0) = 1$.
 $r^2 + 2r - 3 = (r + 3)(r - 1) = 0 \Rightarrow r = -3, 1$. So $y = C_1e^{-3x} + C_2e^x$. From $y(0) = 1$: $C_1 + C_2 = 1$. From $y'(0) = 1$: $-3C_1 + C_2 = 1$. Subtracting gives $4C_1 = 0$, so $C_1 = 0$, $C_2 = 1$. Thus $y = e^x$.

Repeated roots

- 5 Solve $y'' - 6y' + 9y = 0$.
 $r^2 - 6r + 9 = (r - 3)^2 = 0 \Rightarrow r = 3$ (repeated). General solution: $y = (C_1 + C_2x)e^{3x}$.
- 6 Solve $y'' + 4y' + 4y = 0$, $y(0) = 2$, $y'(0) = -1$.
 $r^2 + 4r + 4 = (r + 2)^2 = 0 \Rightarrow r = -2$. So $y = (C_1 + C_2x)e^{-2x}$. From $y(0) = 2$: $C_1 = 2$. Differentiating, $y' = C_2e^{-2x} - 2(C_1 + C_2x)e^{-2x}$; $y'(0) = C_2 - 2C_1 = -1 \Rightarrow C_2 = 2(2) - 1 = 3$. Thus $y = (2 + 3x)e^{-2x}$.

Complex roots

- 7 Solve $y'' + 4y = 0$.
 $r^2 + 4 = 0 \Rightarrow r = \pm 2i$. General solution: $y = C_1\cos 2x + C_2\sin 2x$.
- 8 Solve $y'' - 2y' + 5y = 0$.
 $r^2 - 2r + 5 = 0 \Rightarrow r = \frac{2 \pm \sqrt{4 - 20}}{2} = 1 \pm 2i$. General solution: $y = e^x(C_1\cos 2x + C_2\sin 2x)$.

Application: a damped spring

- 9 A mass–spring–damper system satisfies $y'' + 4y' + 5y = 0$. Find the general solution and classify the damping (over/critically/under-damped).

$r^2 + 4r + 5 = 0$, discriminant $= 16 - 20 = -4 < 0$, so $r = -2 \pm i$. General solution:

$y = e^{-2x}(C_1 \cos x + C_2 \sin x)$ — this is under-damped (complex roots with negative real part give oscillatory decay).

- 10 Solve $y'' + 6y' + 9y = 0$, $y(0) = 1$, $y'(0) = 0$, and classify the damping.

$r^2 + 6r + 9 = (r + 3)^2 = 0 \Rightarrow r = -3$ (repeated) — critically damped. So $y = (C_1 + C_2 x)e^{-3x}$. From $y(0) = 1$: $C_1 = 1$. $y' = C_2 e^{-3x} - 3(C_1 + C_2 x)e^{-3x}$; $y'(0) = C_2 - 3C_1 = 0 \Rightarrow C_2 = 3$. Thus $y = (1 + 3x)e^{-3x}$.